

Final Project Proposal:
Prior Sensitivity in Bayesian Logistic Regression:
A Small-Sample Medical Study

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Introduction

This project applies Bayesian logistic regression to a small medical dataset to examine how prior choice shapes posterior inference. A fundamental property of Bayesian inference is that the influence of the prior diminishes as sample size grows. In small clinical studies, common in medicine due to cost and ethical constraints, prior choice can substantially alter conclusions. Understanding this dependence is critical for practitioners who must choose priors carefully and communicate uncertainty honestly.

Dataset

I will use the `birthwt` dataset ($n = 189$) from R's `MASS` package, collected at Baystate Medical Center, Springfield, MA (1986). The binary outcome is low birth weight (< 2.5 kg), and predictors include maternal age, weight, race, smoking status during pregnancy, history of previous premature labour (`ptd`), history of hypertension, uterine irritability, and number of physician visits during the first trimester.

Methods

Two analyses are planned:

1. **Prior Sensitivity Analysis** (full data, $n = 189$): Fit Bayesian logistic regression under three priors: diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and clinically informative priors motivated by medical literature (e.g., positive coefficients for smoking and hypertension). Compare posterior means and 95% credible intervals across all three models.
2. **Subsampling Experiment**: Repeatedly subsample at $n \in \{20, 40, 80, 189\}$ and refit all three models. Visualize how posterior distributions and credible interval widths evolve with sample size, illustrating the transition from prior-dominated to likelihood-dominated inference.

All analyses will be conducted in R using `rstanarm` and visualized with `bayesplot` and `ggplot2`.

Goals

- Demonstrate how prior sensitivity varies with sample size in a real medical context.
- Identify which risk factors (e.g., smoking, hypertension) are most affected by prior choice under small n .
- Provide practical guidance on prior selection for small clinical datasets, contrasting diffuse, weakly informative, and domain-informed priors.